Open IC leads after 2026-10-11 — search-round report

Source: `research/ic_open_leads_scoping_20261011/REPORT.md` (aburan28/crypto). Each statement is labelled **measured**, **derived** (calculated from recorded values or formulas) or **scoped** (from the repo's own notes, not run).

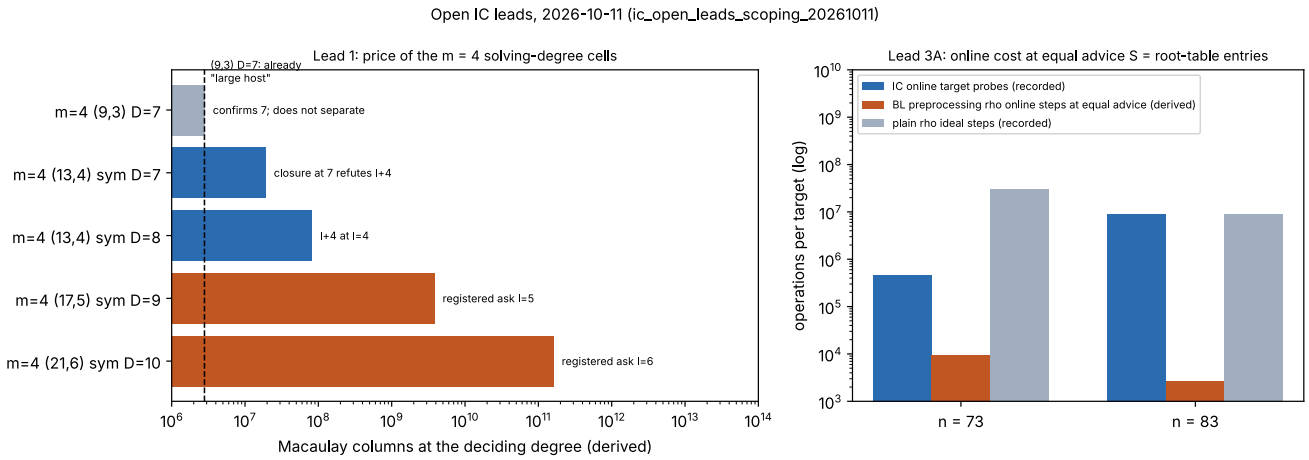


Figure 1: Left: Macaulay columns at the deciding degree for the m = 4 solving-degree cells (derived, $\Sigma\, C(N, k \le D)$); the dashed line is the (9,3) degree-7 matrix the DREG note already classes as a large-host job. Right: online operations per target at equal advice on the n = 73 / 83 fixtures — IC target probes and plain-rho ideal steps recorded, BL online steps derived. Data: `reports/data.json`.

lead	status	evidence
1. m = 4 solving degree, ℓ = 4–6	scoped: at $\ell \le 3$ "$\ell + 4$" and "levels at 7" predict the same D; the first separating m = 4 cell is $\ell = 4$ (8 vs 7), $\approx 1.9 \cdot 10^7$ columns at D = 7, $8.1 \cdot 10^7$ at D = 8. Cheapest decisive cell in the programme: m = 3 (15,5) at D = 7 on a ≈ 128 GB host.	derived
2. equivariant F4 ⟶ full F4	scoped: rank-only probe at m = 2 today; full F4 is 3–5 days and needs a Frobenius-invariant m = 4 formulation first; constant ($\le$ n) lever.	scoped
3A. preprocessing rho at equal advice	IC online loses ≈ 48× (n = 73) and ≈ 3,400× (n = 83) in operations; the note's n = 73 figure used rank-stage probes; BL precompute is $\approx 10^{11}$ steps, not "seconds".	derived
3B. free-relation census	rank 49 / 1 / 6 against 599 needed at n = 61 / 73 / 83; note's count high by ≈ n; rank stage not free.	measured
4. D_4-symmetrised S_5 lattice	1/80 is the Jochemsz–May bound, never measured; e-rewrite and $(Z/2)^3$ invariants leave the determinant condition unchanged; 13× exponent gap; closed by counting.	derived
5. tree split round 2	pre-registered, run; see its RESULTS.md and run report.	measured

Recommendation. The one remaining exponent-relevant experiment is a solving degree: run m = 3 (15,5) at degree 7 on a 128–256 GB host (pre-registered, existing `dreg_ladder`); only if it levels at 7, build the m = 4 barrel formulation plus equivariant blocks and test (13,4) closure at degree 7. Nothing in this round supports an exponent, speedup or deployed-curve claim.